7CS039 Statistics for AI & Data Science

Week 1 Descriptive Statistics

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1 Introduction

Statistics is: "the science of making effective use of numerical data relating to groups of individuals or experiments. It deals with all aspects of this, including not only the collection, analysis and interpretation of such data, but also the planning of the collection of data, in terms of the design of surveys and experiments". This is as good a definition as any but it's also extremely broad. For our purposes we will be concerned with employing/using statistics. Employing statistics serves two purposes,

- **Description**: statistics is used to describe the characteristics of groups. These characteristics are referred to as *variables*. Data is gathered and recorded for each variable. Descriptive statistics can then be used to "say things" about each variable. For instance, if we are concerned with a group of students, possible variables, are their ages, their sex, their heights, their level of fitness etc. Possible descriptive statistics are that the mean height of the students is 1.65 metres, the maximum age of the male students is 37, or 46% of the students are male. Most people would consider diagrams, for example, a histogram of the ages of the students, as coming under the general heading of "descriptive statistics"

- **Inference** (prediction): Statistics is also frequently used for purposes of inference. Inference is based on the concept of generalisability: if enough data is compiled about a particular context (e.g., the heights of a given group of students), the patterns revealed through analysis of the data collected about that context can be generalized to (or predicted to occur in) similar contexts. For instance, I am willing to predict that the average height of the males in this class is greater than that of the females. Why? I know from my experience that it is (nearly always) the case that in a randomly selected group of adults the average height of the males will be greater than the females. Whilst you have not been randomly selected, your height has nothing to do with how you were selected, and thus it seems reasonable to infer from my knowledge of the heights of males and females that the average height of the males doing this course will be greater than that of the females.

  Question: What if this course was being taught to 13 year olds?

  Similarly, if you were to take a "random sample" of $10,000$ people who watch a given television programme, and find that $6,000$ of them were female, you would probably feel happy to infer that, in general, more women than men watch that programme. (On the other hand if you were to take a random sample of 10 people and find that,

out of those 10, 6 were female you probably would *not* feel comfortable inferring that, in general, more women than men watch that program, even though in both cases 60% of women and 40% of men said that they watched the programme. Later on in this course we will see why this is so.)

2 Descriptive Statistics: Measurement, Location and Dispersion

2.1 Nominal/Categorical, Ordinal and Interval Scales

Data is central to statistics. A common reply to the question "what is data" is that it is "numbers to do with something". Whilst this is not strictly true, usually data is in the form of numbers. It is therefore important to consider the nature of the numbers being used in any particular set of data, as we see in this section, the number "3" for example is not always the same.

2.2 Nominal/Categorical scale

At the nominal scale, i.e., for a nominal category, one uses labels; for example, people may be male or female, rocks can be generally categorized as igneous, sedimentary and metamorphic. Often numbers are used to code nominal variables, for example: *male =1*, female=2; *igneous, =1*, *sedimentary=2*, *metamorphic=3*. What is important is that the numbers are being used as labels, they are not quantities of any kind. For instance, if you had one igneous rock ($= 1$) and one metamorphic rock ($= 3$), clearly it makes no sense to say that as $\frac{1+3}{2} = 2$, "on average" you have a sedimentary rock! Variables assessed on a nominal scale are also called **categorical** variables.

2.3 Ordinal scale

In this scale type, the numbers assigned to objects or events represent the rank order (1st, 2nd, 3rd etc.) of the entities assessed, but do not describe quantity. An example of ordinal measurement is when a book is ranked 1 star, 2 star etc. If some one ranks one book $\star$ and another $\star\star\star$ the person certainly thinks the second book is better than the first, but not necessarily three times as good.

2.4 Interval/Quantitative scale

Quantitative (i.e. those that express quantity) attributes are all measurable on interval scales, as any difference between the levels of an attribute can be multiplied by any real variables that number to exceed or equal another difference. Thus *height, weight, temperature, distance, family size, age* are all measured on the interval scale. It is important to be aware of differences that exist between interval scale variables. For instance, if you were $30cms$ long when you were born, and are $180cms$ tall now, then you have been every single height between $30cms$ and $180cms$ at some stage in you life, i.e. there was a time you were $111.1cms$ tall, a time you were $10\pi cms$ tall etc. If you consider family size

(which we shall take to mean how many brothers and sisters you have, plus yourself) this is still a quantity, but is different to height in as far as, for instance, say you are the eldest of six children, when you were born your family size was one, when your next sibling was born the family size "jumped" to two, it then moved to three, four etc. but there was never a time when your family size was $3\frac{2}{9}$. We may therefore subdivide interval data into *continuous* and *discrete* data.

2.4.1 Continuous Data

Interval data is said to be continuous if the values belonging to it may take on *any* value within a finite or infinite interval. You can count, order and measure continuous data.

2.4.2 Discrete Data

Interval data is said to be discrete if the values belonging to it are distinct and separate, i.e. they can be counted $(1, 2, 3, \ldots)$. Examples might include the number of kittens in a litter; the number of patients in a doctor's surgery; the number of flaws in one metre of cloth. Note that it is not necessarily true that discrete data can only take on integer values, if data could only take values 3, $3\frac{1}{4}$, $3\frac{1}{2}$, $3\frac{3}{4}$ and 4, and these were measurements of some sort, then this would be discrete data. The distinguishing feature is that there are "gaps" between any two possible values.

3 Measures of Central Tendency ("Averages")

The purpose of descriptive statistics is to describe data. If we were to ask people to describe the age of a certain group of people we would hear statements such as:

- On average they are twenty.
- Half of them are over twenty-one.
- More of them are eighteen than any other age.

These three descriptions correspond to the three most common measures of central tendency, the *mean*, *median* and *mode*.

3.1 Mean

The *mean* (or, more correctly, the *arithmetic* mean) of n numbers $x_1, x_2, \ldots, x_n$ is given by the formula:

$$\mu = \frac{\sum x_i}{n} \quad or \quad \bar{x} = \frac{\sum x_i}{n}$$

where

- $\mu =$ mean of a population.

- $\bar{x}$ =mean of a sample.

In layperson's language: we add up all the numbers and divide them by the number of numbers that we have.

Example

The mean of $\{1, 3, 6, 2\}$ is:

$$\frac{1+3+6+2}{4} = 3$$

3.2 Median ... "The one in the middle"

To calculate the median of a set of data we:

- Order the data. (either from biggest to smallest or vice-versa, it does not matter).
- Find the item of data above which half the data lies, (and hence below which half the data lies).

Example

Find the median of $\{2, 7, 3, 1, 0, 4, 2\}$

Solution

- Order the data: $0, 1, 2, \overset{\downarrow}{2}, 3, 4, 7$
- Clearly half the data lies above 2 and half below it.

Hence the median $= 2$.

The above only works if there is an odd number of numbers in the set being considered. If there is an even number we take the mean of the two numbers in the middle.

Example

Find the median of $2, 7, 3, 0, 1, 0, 1, 5 =?$

Solution

1. Order the data: $0, 0, 1, \overset{\downarrow}{1}, \overset{\downarrow}{2}, 3, 5, 7$
2. Median $= \frac{1+2}{2} = \frac{3}{2}$ or 1.5

Note that half the data lies above 1.5 and half below 1.5

Note that if there are n items of data in the set being considered then the median is the item of data in $\dfrac{n+1}{2}$ position (after ordering).

3.3 Mode

The mode of a set of data is the data value that occurs most frequently.

Example

The mode of: $\{0, 1, 3, 2, 2, 1, 0, 2\}$ is 2.

Example

The mode of: $\{0, 0, 1001, 1002, 1003, 1004, 1006, 1007, 1008, 1009, 1010\}$ is 0.

If two observations are equally frequent we say that the data is *bimodal*. If no data value occurs (at least) twice then the data does not have a mode. Remember that the mode is an attempt (as are the mean and median) to use one number to describe a whole set of data. Often the mode does not succeed very well at this task. For instance if I were to ask any group of people how many units of alcohol they drank in the last week, the mode will almost certainly be zero, as more people will not have drunk at all than will have drunk any given positive number of units.

4 Other measures of Location

4.1 The Maximum and Minimum

These are self explanatory, and simply represent the greatest and least data values.

4.2 Q_1 and Q_3, The Upper and Lower Quartiles

The Lower Quartile may be roughly defined as the value that is a "quarter of the way up" the data. Again, different people have slightly different ways of determining exactly what this value is; usually however all methods lead to approximately the same answer. We shall use the definition that the lower quartile is the item in $\frac{n+1}{4}$ position. Similarly the upper quartile may be roughly defined as the value that is a "three quarters of the way up" the data. We shall use the definition that the lower quartile is the item in $\frac{3(n+1)}{4}$ position. Note that the upper quartile is sometimes denoted by Q_3, and the lower quartile by Q_1, Q_2 being the median.

Example

Find the median and upper and lower quartiles of $\{1, 4, 2, -2, 8, 10, 3, 6, 3\}$
Solution:

- Order the data from lowest to highest: $\{-2, 1, 2, 3, \overset{\downarrow}{3}, 4, 6, 8, 10\}$
- Here $n = 9$, so:
- the median position is the data value in $\frac{9+1}{2} = 5^{th}$ position, i.e. 3.
- the lower quartile (Q_1) is the data value in $\frac{9+1}{4} = 2.5^{th}$ position, i.e. 1.5.
- the upper quartile (Q_3) is the data value in $\frac{30}{4} = 7.5^{th}$ position, i.e. 7.0.

Example

Find the median, upper and lower quartiles of:

5	11	7	8	3	11	8	10	9	7	5	5
6	11	6	6	10	8	14	13	6	5	8	2

Solution:

- Sort the data:

2	3	5	5	5	5	6	6	6	6	7	7
8	8	8	8	9	10	10	11	11	11	13	14

- Here $n = 24$, so the median position is $\frac{24+1}{2} = 12.5^{th}$ position, i.e. 7.5
- The lower quartile (Q_1) is the data value in $\frac{24+1}{4} = 6.25^{th}$ position, i.e. a quarter of the way between the 6th and 7th items of data, i.e. 5.25.
- The upper quartile (Q_3) is the data value in $\frac{3(24+1)}{4} = 18.75^{th}$ position, i.e. a three-quarters of the way between the 18th and 19th items of data, i.e. 10.00.

Note that other approaches to finding Q_1 and Q_3 may result in slightly different values.

5 The Five Number Summary

The *Five Number Summary* consists of the minimum, lower quartile, median, upper quartile and maximum of a set of data. i.e.

$$\{\text{Min},\ Q_1,\ \text{Median},\ Q_3,\ \text{Max}\}$$

These five statistics taken together will give a better description of the data than any one of them on its own. For example, the five number summary for the data

5	11	7	8	3	11	8	10	9	7	5	5
6	11	6	6	10	8	14	13	6	5	8	2

presented previously is:

$$\{2, 5.25, 7.5, 10, 14\}$$

6 Measures of Spread/Dispersion

Without knowing something about how data is dispersed, measures of central tendency may be misleading. For example, a residential street with 20 homes on it having a mean value of £200,000 with little variation from the mean would be very different from a street with the same mean home value but with 3 homes having a value of 1 million and the other 17 clustered around £60,000. Measures of dispersion provide a more complete picture. Dispersion measures include the range, mean absolute deviation, interquartile range, variance and standard deviation. Measures of dispersion are also sometimes known as *measures of spread*.

6.1 The Range

Probably the simplest measure of dispersion is the *range*. It is calculated by simply taking the difference between the maximum and minimum values in the data set.

Example

For example, if $D = \{4, 5, 2, 4, 6, 6, 10, 8, 4, 6, 7, 6\}$, then

$$Range(D) = 10 - 2 = 8$$

The range only provides information about the maximum and minimum values and does not say anything about the values in between. It is however easy to calculate, and widely used. Note that it is very sensitive extreme values. For instance if we add a single additional entry of 100 to D in the example above, then the range increases to 98.

6.2 The Interquartile Range

The *interquartile range* (IQR) of data Y is simply the difference between the upper and lower quartiles of Y, i.e.

$$IQR = Q_3 - Q_1$$

As the values of the largest and smallest observations are not involved in its calculation, it is not influenced by extreme data values. It is the range of the "middle half" of the data, and hence is a very useful measure of dispersion if one is interested only in "typical cases".

6.3 Variance and Standard Deviation

The most commonly used measure of dispersion is the *standard deviation*, which is the square root of the *variance*. The variance of a data set is calculated by taking the mean of the squared differences between each value and the mean value. The calculation and notation of the variance and standard deviation depends on whether we are considering the entire population or a sample set. Following the general convention of using Greek characters to express population parameters and Arabic characters to express sample statistics, the notation for standard deviation and variance is as follows:

- population standard deviation $= \sigma$
- population variance $= \sigma^2$
- estimate of population standard deviation based on sampled data $= s$
- estimate of population variance based on sampled data $= s^2$

Definition 6.1. The population variance of data $\{x_1, x_2, \ldots, x_n\}$ is defined as:

$$\sigma^2 = \frac{\sum(x_i - \mu)^2}{n}$$

where μ is the mean of the data.

Definition 6.2. The population standard deviation is the square root of this value, i.e.

$$\sigma = \sqrt{\frac{\sum(x_i - \mu)^2}{n}}$$

The variance of a sampled subset of observations is calculated in a similar manner, using the appropriate notation for sample mean and number of observations.

Example

The ages of the four children in a pediatric ward are 4, 5, 7 and 8. What is the variance of these ages?

Solution:

Here

- $\mu = \dfrac{4+5+7+8}{4} = 6.$
- $\sigma^2 = \dfrac{(4-6)^2 + (5-6)^2 + (7-6)^2 + (8-6)^2}{4} = \dfrac{4+1+1+4}{4} = 2.5.$

7 Graphical Techniques

In this section we discuss several types of graphs and diagram that are used to represent data. There are numerous graphical techniques and we will only discuss a few of them here.

7.1 Histograms

The most familiar diagram for summarizing data is most probably the *histogram*. The term was introduced by Karl Pearson in 1895.

Example

The frequency table below records the heights of $11,327$ males living in a fictional town.

Height x (cm)	Frequency
$151 \leq x < 154$	4
$154 \leq x < 157$	17
$157 \leq x < 160$	79
$160 \leq x < 163$	229
$163 \leq x < 166$	545
$166 \leq x < 169$	1009
$169 \leq x < 172$	1615
$172 \leq x < 175$	2064
$175 \leq x < 178$	2043
$178 \leq x < 181$	1656
$181 \leq x < 184$	1110
$184 \leq x < 187$	571
$187 \leq x < 190$	259
$190 \leq x < 193$	94
$193 \leq x < 196$	25
$196 \leq x < 199$	5
$199 \leq x < 203$	1

The data is shown in the histogram below

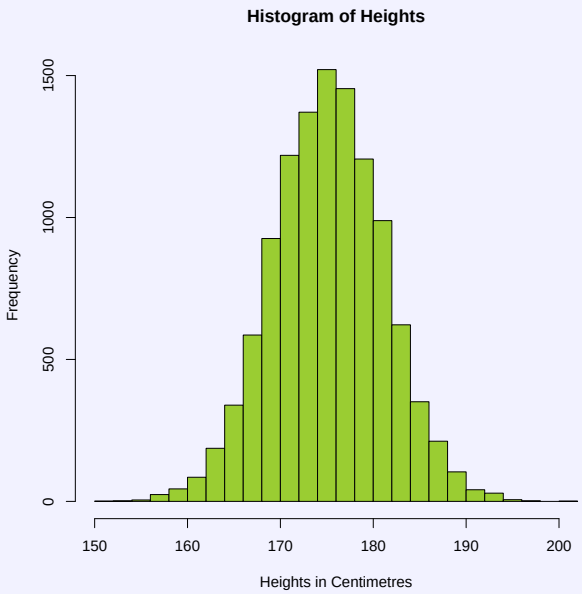


Figure 1: Histogram of Heights

A common misconception exists regarding the purpose of a histogram. Whilst it is

certainly possible to measure the height of the bar representing the interval 169–172 and determine that there are 1,615, or 3.64%, of men in this height range, that is not the principal purpose of the histogram. *The principal purpose of the histogram is to give a quick visual impression of the data.* We see at a glance that the most common height is in-or-about 175cms, and that the heights fall off evenly either side of this. (Later in the course we shall see that this corresponds to what is known as a *normal distribution.*)

7.2 Stem and Leaf Plot

Stem and leaf plots may be regarded as a variation of a histogram that maintains the principal purpose of a histogram in that they provide a quick visual impression of the distribution of the data, but in addition enable the data itself to be retrieved from the plot. They are however only useful if there is not "too much" data. Usually stem and leaf plots are only used with "count data", i.e. data that are whole numbers, but they may be used if it is permissible to round numbers to the nearest integer.

Example

Consider the following dataset on the number of rooms in a hotel occupied each day during the month of June.

55	49	37	57	46	40	64	35	73	62
61	43	72	48	54	69	45	78	46	59
40	58	56	52	49	42	62	53	46	81

The following histogram represents this data.

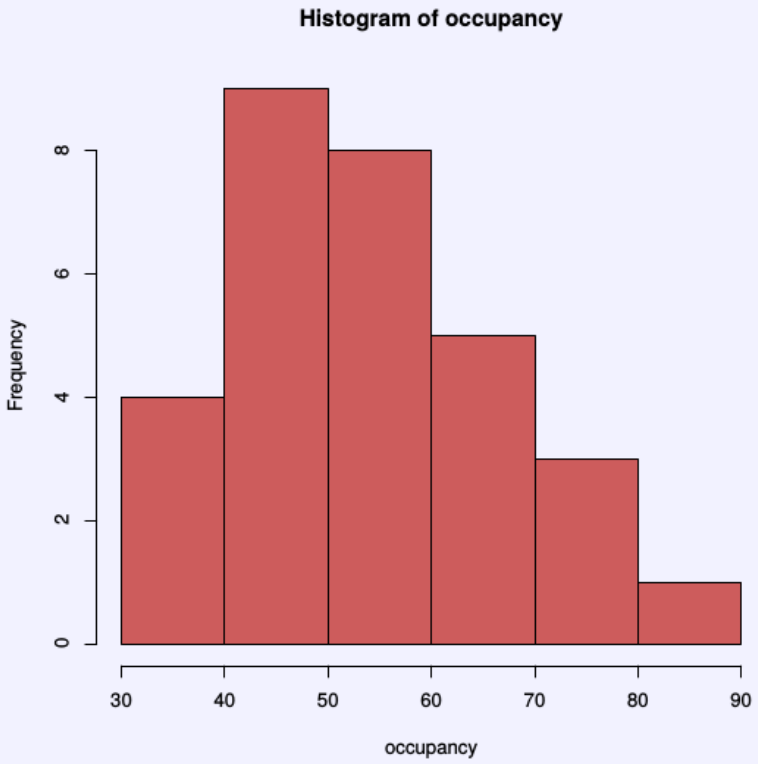


Figure 2: Histogram of Room Occupancy

Whilst the histogram rapidly and accurately shows that the data is *skewed to the right*, i.e. the right hand tail of the data is longer than the left, (more about this later in the course), and we can retrieve the information that their were eight days when between 50 and 60 rooms were occupied, we lose, for example, the information that there was only one day when 56 rooms were occupied.

Example

A stem and leaf diagram for the data, with *stem unit* $= 10$, is as below:

3	5	7								
4	0	0	2	3	5	6	6	6	8	9	9
5	3	3	4	5	6	7	8	9			
6	1	2	2	4	9						
7	2	3	8								
8	1										

Table 1: Stem and leaf Plot of Room Occupancy, stem unit $= 10$

As is apparent, the left-most column represents the intervals $30, 40, \ldots, 80$ with the quantities within these intervals being represented by the "leaves". The stem and leaf plot is like a histogram turned on its side, with the blocks being replaced by the individual numbers. A stem and leaf diagram for the same data, with *stem unit* $= 5$, is:

3	5	7					
4	0	0	2	3			
4	5	6	6	6	8	9	9
5	3	3	4				
5	5	6	7	8	9		
6	1	2	2	4			
6	9						
7	2	3					
7	8						
8	1						

Table 2: Stem and leaf Plot of Room Occupancy, stem unit $= 5$

Note that stem-and-leaf diagrams are only practical if the amount of data being represented is reasonably small.

7.3 Boxplots

A boxplot is a diagram that looks like the below:

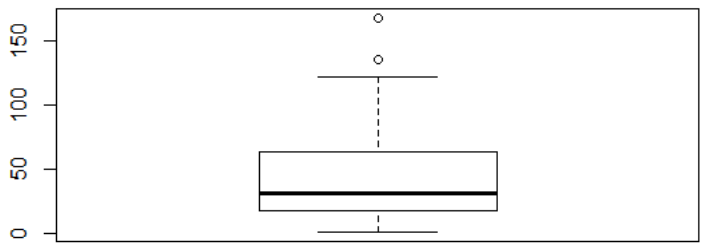


Figure 3: Simple Boxplot

Boxplots concentrate on giving a quick visual interpretation of the distribution of the data utilizing values of the *five number summary*

$$\{min, Q_1, median, Q_3, max\}$$

The steps used in drawing box plots are the following:

- Calculate the median, the upper quartile (Q_3) and lower quartile (Q_1).
- Calculate the interquartile range (IQR) by subtracting the lower quartile from the upper quartile. i.e. $IQR = Q_3 - Q_1$
- Construct a box parallel to the vertical axis with the upper edge corresponding to the value of Q_3 and the lower edge to Q_1.
- Indicate where the median lies inside of the box by drawing a horizontal line.
- Any data observation which lies more than $1.5 \times IQR$ lower than the lower quartile or $1.5 \times IQR$ higher than the upper quartile is considered an outlier. Indicate where the smallest value that is not an outlier is by connecting it to the box with a vertical line or "whisker" and cap this "whisker" with a small horizontal line. Likewise, connect the largest value that is not an outlier to the box by a "whisker" and mark it with a small horizontal line.
- Indicate *outliers* (untypical observations) by a cross/circle.

7.3.1 Multiple Boxplots

Boxplots are particularly useful for comparing the distributions of data that has been classified into several categories.

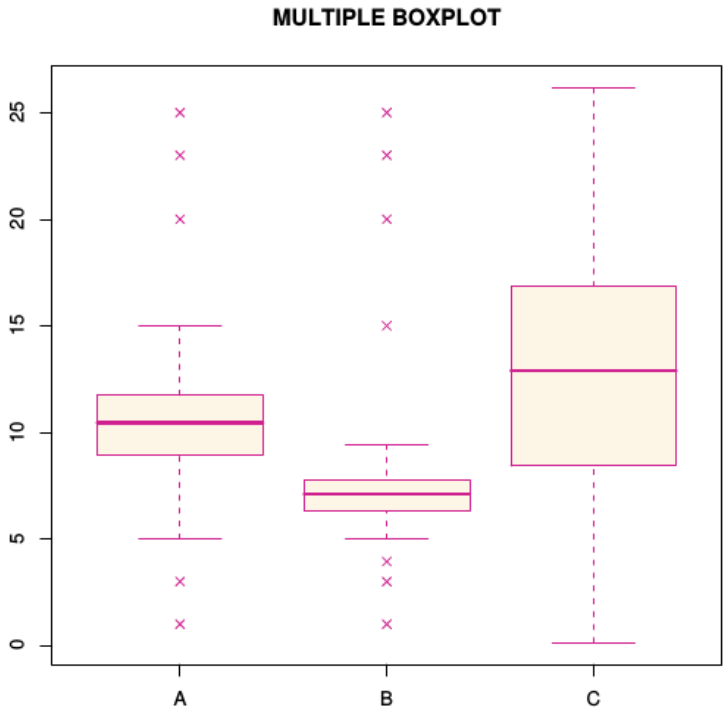


Figure 4: Multiple Boxplot

Multiple boxplots are sometimes referred to as *categorised* boxplots. For instance, the multiple boxplot of Figure 4 might refer to the number of days of sick leave taken by three categories of workers, A, B and C. We see immediately that

- The range of days of absenteeism is approximately the same for all three categories,
- in general, category B workers take the least number of days off sick, followed by category A followed by C,
- the variation of the number of days absenteeism is greatest amongst category C workers, and least for category B.

Example

In an experiment the times taken by people to perform a given task is recorded. The boxplot of Figure 5 illustrates the results of the experiment where the times have been categorised by age-group. ($21 - 30$ etc). Comment upon the results of the experiment.

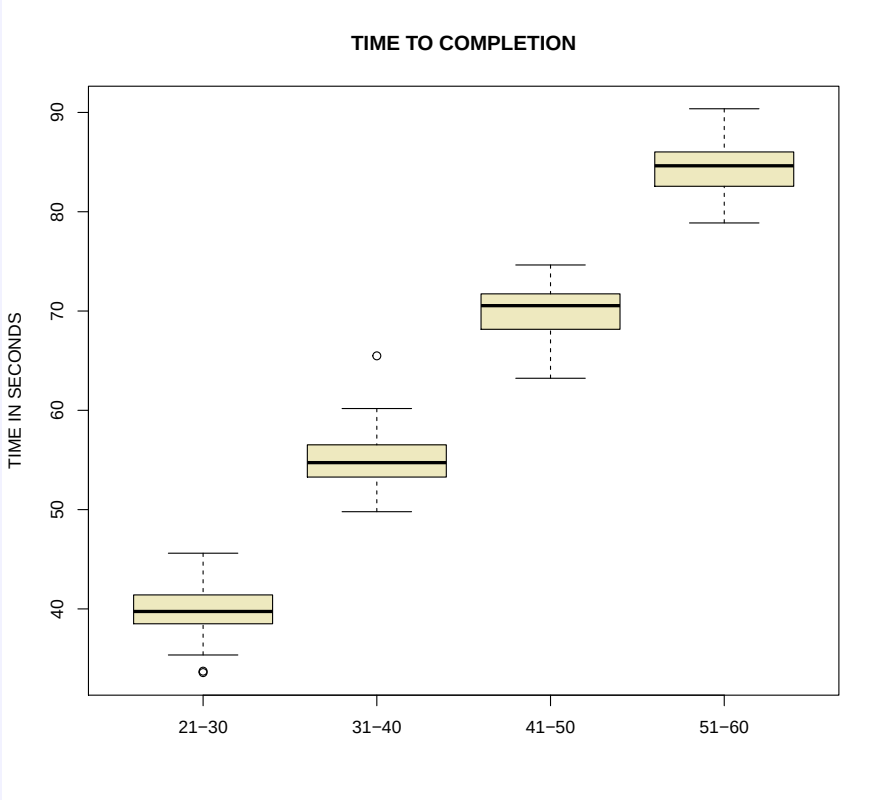
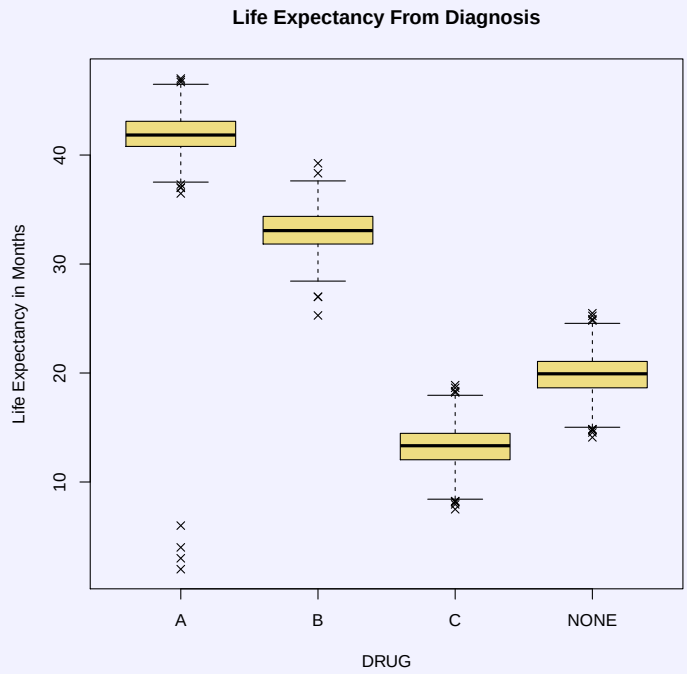


Figure 5: Times taken to perform task

Example

If left untreated, the median life expectancy from diagnosis of a certain type of cancer is 20 months. Three drugs to treat this cancer are each tested on 500 patients, and the time (in months) from diagnosis to death is recorded for each patient. The resultant data are illustrated by the multiple boxplot of Figure 6. The boxplot labeled "None" illustrates the times-to-death of patients that received no treatment.

Figure 6: Multiple (Categorised) Box Plot



What may we conclude?

8 Understanding Statistics

Statistics is primiarily concerned with making valid inferences about populations based on information containmed in samples taken from those populations.

- **Sampling**
  - Sample - "a part to show what the rest is like".
  - Sample - "a portion, part or piece, taken or shown as representat8ive of the whole".

  But, what does the "rest" or the "whole" mean? What is the sample a sampke of? If I observe the hairstyles of the people walking into Tesco today, are my observations

reflective of the hairstyles of all the people who shop in that particular Tesco, all the people who shop in tesco, or all the people who shop? the answer could be any or all of the above.

- Good Samples - lead to meaningful generalizations.
- Bad Samples - lead to complete rubbish!

Question: How should we sample?

- **Estimation**

  We want to estimate the value of some population parameter based on a good sample. The first thing we need is a "good" sample - we'll need to knwo hwo to do that.

  The population may be large (even infinite). How can we use a sample (possible a small one) to say something meaningful about the population? We use a **confidence interval**.

- **Hypothesis Testing**

  We make or are given some claim or conjecture regarding a population. We take a sample, analyse it, and then decide whether to accept or reject the original claim.

- **Regression & Correlation**

  Aim: Predict the value of one variable in terms of the known value of another variable.

  e.g. Future sales of a product in terms of price.

  e.g. A persons weight loss in terms of the number of weeks he/she has been on a new diet.

We'll try and get a feel for how statistics is used in the real world in this module - the emphasis will be on using statsitics as a tool. We will provide opportuhjites for deeper study for thso who are interested.

9 Sampling & The Sampling Distribution of the Mean

Suppose we have a sample of n people from athe population and we record the height of these n people. The mean height of the sample is

$$\overline{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

Question: Is the sample mean $\overline{x}$ "close to" the population mean μ?

Answer: We don't know unless we take the mean of the entire population.
However, we can say "how likely" we are to get a sample mean close to the population mean. We could take all possible samples of size n, find the mean of each sample, and thus get a distribution. Let

$$\overline{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

be the mean of a random sample. This is a random variable (i.e. not a fixed number) and it has a distrbution called the **sampling distribution of the mean**.

Example

Suppose the population is the fiite set $\{3, 5, 7, 9, 11\}$ of size $N = 5$. We will construct the sampling distruibution of the mean for a sample of size $n = 2$ drawn without replacement from the population. The population mean is

$$\mu = \frac{3+5+7+9+11}{5} = 7$$

and the population standard deviation is

$$\begin{aligned}\sigma &= \sqrt{\frac{(3-7)^2+(5-7)^2+(7-7)^2+(9-7)^2+(11-7)^2}{5}} \\ &= \sqrt{8} \approxeq 2.828\end{aligned}$$

If we take random sampkles of size $n = 2$ from the population there are ten ways to do this.

$$\begin{array}{ll} (3,5) & \overline{x} = 4 \\ (3,7) & \overline{x} = 5 \\ (3,9) & \overline{x} = 6 \\ (3,11) & \overline{x} = 7 \\ (5,7) & \overline{x} = 6 \\ (5,9) & \overline{x} = 7 \\ (5,11) & \overline{x} = 8 \\ (7,9) & \overline{x} = 8 \\ (7,11) & \overline{x} = 9 \\ (9,11) & \overline{x} = 10 \end{array}$$

Sinjxce we are taking samplea at random each ofthese ten samples is equally likely and so the sample means are equally likley with probability $\frac{1}{10}$. This gives us the following "sampling distrubution of the mean".

$\overline{x}$	Probability
4	$\frac{1}{10}$
5	$\frac{1}{10}$
6	$\frac{2}{10}$
7	$\frac{2}{10}$
8	$\frac{2}{10}$
9	$\frac{1}{10}$
10	$\frac{1}{10}$

Now we can see that

- The probability that a sample mean will differ from the population mean ($\mu = 7$) by 1 or less is $\frac{6}{10}$.
- The probability that a sample mean will differ from the population mean ($\mu = 7$) by 2 or less is $\frac{8}{10}$.

So, if we didn;t know the population mean, and wanteds to estimate it with a random sample of szie $n = 2$, this would give us some idea about the potential size of our error.

In the last example we could find more informationm about the sampling distruibution fo the mean by calculating its mean $\mu_{\overline{x}}$ and its standard deviation $\sigma_{\overline{x}}$. Here we would find

$$\begin{aligned}
\mu_{\overline{x}} &= \left(4 \times \frac{1}{10}\right) + \left(5 \times \frac{1}{10}\right) + \left(6 \times \frac{2}{10}\right) + \left(7 \times \frac{2}{10}\right) + \left(8 \times \frac{2}{10}\right) + \left(9 \times \frac{1}{10}\right) + \left(10 \times \frac{1}{10}\right) \\
\sigma_{\overline{x}} &= (4-7)^2.\frac{1}{10} + (5-7)^2.\frac{1}{10} + (6-7)^2.\frac{2}{10} + (7-7)^2.\frac{2}{10} + (8-7)^2.\frac{2}{10} + (9-7)^2.\frac{1}{10} \\
&+ (10-7)^2.\frac{1}{10} \\
&= 3
\end{aligned}$$

Observe that in this sample

$$\begin{aligned}
\mu_{\overline{x}} &= \mu \\
\sigma_{\overline{x}} &<^{\sigma}
\end{aligned}$$